

Linked List Question Bank

Section A: Multiple Choice Questions (1-15)

1. What is the first node of a linked list called? a) Root b) Head c) Tail d) Origin **Answer: b**

2. What does the

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next
```

pointer of the last node in a singly linked list point to? a) The head b) Itself c) null d) A random node **Answer: c**

3. What is the time complexity to access the k-th element in a singly linked list? a) $O(1)$ b) $O(\log n)$ c) $O(n)$ d) $O(n^2)$ **Answer: c**

4. Which type of linked list allows traversal in both directions? a) Singly Linked List b) Doubly Linked List c) Circular Linked List d) Static Array **Answer: b**

5. In a circular linked list, the last node's

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next
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points to: a) null b) The previous node c) The head node d) A new empty node **Answer: c**

6. What is the time complexity of inserting a node at the head of a singly linked list? a) $O(1)$ b) $O(n)$ c) $O(\log n)$ d) $O(n^2)$ **Answer: a**

7. In Java, what is stored inside a

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Node
```

object of a singly linked list? a) Only data b) Only a pointer c) Data and a reference to the next node d) Data and two references **Answer: c**

8. Which Java class provides a built-in, ready-to-use Linked List implementation? a) `java.util.ArrayList` b) `java.util.LinkedList` c) `java.util.HashMap` d) `java.util.Stack` **Answer: b**

9. What happens to a deleted node in Java's manually implemented linked list? a) It's immediately erased from RAM by the programmer b) It becomes unreachable and is cleaned up by the Garbage Collector c) It automatically becomes the new head d) Nothing, it stays in the list **Answer: b**

10. What is a major disadvantage of a Doubly Linked List compared to a Singly Linked List? a) Cannot be traversed forward b) Uses more memory per node c) Cannot store integers d) Cannot be searched **Answer: b**

11. Which of these is a real-world application of a Circular Linked List? a) Binary search b) Round-robin CPU scheduling c) Sorting an array d) Hashing **Answer: b**

12. What is the risk of using

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while(current != null)
```

to traverse a Circular Linked List? a) It will only print one element b) It will throw a compile-time error c) It can result in an infinite loop d) It will sort the list **Answer: c**

13. What is the space complexity of a singly linked list with n nodes? a) $O(1)$ b) $O(\log n)$ c) $O(n)$ d) $O(n^2)$ **Answer: c**

14. Which pointer(s) does a node in a Doubly Linked List contain? a) Only next b) Only prev c) next and prev d) head and tail **Answer: c**

15. Why is inserting at the end of a singly linked list (without a tail pointer) $O(n)$? a) Because Java is slow b) Because you must traverse from head to find the last node c) Because arrays are faster d) Because nodes are sorted **Answer: b**

Section B: True / False (16-20)

16. A linked list stores its elements in contiguous memory locations, just like an array. **(False)**

17. In a singly linked list, you can move backward from a node to its previous node directly. **(False)**

18. Java's

```
LinkedList
```

class is internally implemented as a doubly linked list. **(True)**

19. Deleting the head node of a singly linked list is an $O(1)$ operation. **(True)**

20. A circular linked list always has a node whose

next

pointer is

null

. (False)

Section C: Fill in the Blanks (21-25)

21. The two fields inside a singly linked list

Node

class in Java are

and

. (Answer: data, next)

22. A linked list where the last node points back to the first node is called a

linked list. (Answer: Circular)

23. In a Doubly Linked List, the pointer that references the previous node is called

. (Answer: prev)

24. Removing a node from a linked list by redirecting the previous node's pointer is called

the node. (Answer: bypassing)

25. The time complexity to search for a value in an unsorted singly linked list is

. (Answer: $O(n)$)

Section D: Short Answer Questions (26-30)

26. Write the Java code for a

Node

class used in a singly linked list.

27. Explain, in your own words, the difference between a Singly Linked List and a Doubly Linked List.

28. Write a Java method

insertAtHead(int data)

that inserts a new node at the beginning of a singly linked list.

29. List three real-world applications of linked lists.

30. Explain why arrays provide $O(1)$ access while linked lists provide only $O(n)$ access, with reference to how each stores data in memory.

Section E: Long Answer / Coding Questions (31-35)

31. Write complete Java code for a singly linked list class with methods to insert at the end, delete by value, and print the list.

32. Compare Array, Singly Linked List, and Doubly Linked List across memory usage, insertion cost, deletion cost, and access time, presenting your answer as a table.

33. Dry-run the following operations on an initially empty singly linked list and draw the resulting list after each step:

insertAtHead(10)

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insertAtHead(20)
```

```
insertAtEnd(30)
```

```
delete(10)
```

34. Explain why a naive traversal loop can cause an infinite loop on a Circular Linked List, and write a corrected traversal method that avoids this issue.

35. Describe a scenario where using a Linked List would clearly be better than using an Array, and a scenario where an Array would be better than a Linked List. Justify both answers.

Answer Key Summary (MCQ + T/F)

Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	b	6	a	11	b	16	F
2	c	7	c	12	c	17	F
3	c	8	b	13	c	18	T
4	b	9	b	14	c	19	T
5	c	10	b	15	b	20	F